Fiscal Year:	FY 2016	**Task Last Updated:**	FY 11/03/2015
PI Name:	Law, Chung K. Ph.D.		
Project Title:	Structure and Response of Spherical Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	11/25/2014	**End Date:**	11/24/2016
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No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Tse, Stephen Ph.D. (Rutgers University)		
Grant/Contract No.:	NNX15AC79A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	NOTE: Continuation of Structure and Response of Spherical Diffusion Flames--NNX09AR03A The effort was divided into experimental and theoretical elements. The experimental effort consisted of using a porous spherical burner to issue fuel mixtures into an oxidizing environment. Different mixtures of fuel species permitted investigation of the phenomena of interest. The experiment effort was conducted primarily in the drop towers at the Glenn Research Center (GRC) and the NASA reduced-gravity aircraft. Space flight experiment was fully defined and approved during the program period. The theoretical effort consisted of the development of computational codes simulating the combustion phenomena of interest, and interpreting the calculated results on the basis of fundamental flame physics. The purpose of this flight definition task is to gain fundamental understanding of the structure and dynamics of diffusion flames in a simple, well-defined flow field. Using a spherical geometry this work considers: 1) gas-phase transient processes, including fuel vapor accumulation inside the flame and far-field diffusion, and their influence on flame movement and the achievement of steady flame behavior; 2) extinction boundaries of steady or		

	near-steady flames and the identification of the associated chemistry at the global, reduced, and detailed levels; 3) the characteristics of partially-premixed flames; 4) flamefront pulsation and cellular instability especially near the extinction limits; 5) the dynamics of rotating flames; and 6) the behavior and structure of quasi-spherical flames in partial gravity environments. These studies are intended to significantly enhance the understanding of diffusion flames, especially providing capability for predicting flame behavior under varied conditions and configurations. As such, this work should contribute to engineering approaches to practical combustion systems, achieving increased energy conversion efficiencies and reductions in unwanted combustion emissions in both terrestrial and extraterrestrial environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Enhanced understanding of flame-flow vortical interaction could lead to improved burner design and operation in terms of energy efficiency and pollutant emissions.
Task Progress:	During the reporting period, results from the following investigations have been obtained. (1) Premixed Cool Flames in the Counterflow. Cool flames are essential features of chemical kinetics of large hydrocarbon fuels and are closely related to the negative temperature coefficient (NTC) phenomenon and engine knock. In this work, the coupling of cool flame chemistry and convective-diffusive transport is numerically and experimentally investigated for premixed flames in the counterflow, with heated N2 flowing against DME/O2/N2 mixture. It is found that low-temperature heat release is suppressed with high strain rate (short residence time), resulting in a stretched low-temperature S-curve response without a cool flame, and that the low-temperature chemistry induced ignition could transition to a high-temperature flame with low enough strain rate (long residence time). A steady-state premixed cool flame exists only with residence time in a strain rate window. Strain rate, dilution, and equivalence ratio effects on premixed cool flame characteristics were further identified. A symmetric counterflow configuration is then simulated to demonstrate the cool flame temperature and flame speed at fixed local strain rate, showing very different controlling chemistry and characteristics compared with the normal laminar flames governed by high-temperature chemistry. In addition, in a companion experimental investigation, premixed cool flames in counterflow were observed with a CCD camera with high sensitivity in the UV spectrum, corresponding to the characteristic wavelength of excited HCHO. The chemiluminescence from the cool flame is found to become more intense with increasing equivalence ratio, and the position of a cool flame is insensitive to variation in the equivalence ratio at the same flow rate. These observations qualitatively agree with the numerical simulations, demonstrating essential features of a new class of flames - the premixed cool flames. (2) On Colliding Spherical Flames: Morphology, Corner Dynamics, and Flame-generated Vorticity. The collision of two identical spherical flames was studied for its relevance in flame interactions in various practical situations including those within turbulent flames. Experimentally, premixtures of fuel (H2, C3H8) and oxidizer were ignited by two synchronized sparks in a closed chamber under atmospheric pressure, with the subsequent flame-front evolution tracked by a Schlieren system with high-speed camera, as well as computationally simulated. Results show that development of the morphology of flame collision can be distinguished in three stages: independent propagation, strong interaction with flattened interfaces, and corner dynamics after merging of the interfaces. It is further demonstrated that propagation of the flame corner is controlled by the interacting mechanisms of preferential diffusion and kinematic restoration, that vorticity of considerable strength is generated downstream of the corner flame segment, and that the kinetic energy of the flame-generated vorticity is substantial as compared to those of the Kolmogorov eddies such that it could have strong influence on the turbulence energy spectrum and flame burning rate. (3) Computational Simulation of Spherical Diffusion Flame Dynamics and Extinction. For the spherical diffusion flame experiments in microgravity slated to be conducted on the space station, numerical simulations have been conducted to refine the test matrix. The basic flame configuration is the spherically-symmetric diffusion flame generated by issuing a fuel mixture into a quiescent oxidizing environment. Upon ignition, the flame is initially located close to the burner surface, subsequently moving outward to reach the steady-state location. H2/CH4/N2 and H2/CH4/He mixtures are to be examined in the test matrix. Therefore, to guide the experiments, computational simulations are performed with detailed chemistry and transport, along with optically-thick and optically-thin radiation models, as well as with no radiative loss. Such flames allow for the investigation of transient flame extinction at the two limits, i.e., radiative and kinetic. A high flow rate can result in a maximum system extinction Damkholer number (Da) beyond which burning is not possible, caused by the reduction in the flame temperature due to radiative heat loss. By transitioning from a moderate flow rate (for which a diffusion flame can be established) to a low flow rate, a minimum system extinction Da may be reached, corresponding to the purely kinetic limit of burning. The dual diffusion- flame extinction modes for specific mixture compositions and initial ambient oxygen concentrations are obtained numerically. For kinetic extinctions, a certain flow rate is applied to ignite and establish a quasi-steady flame, and then the flow rate is abruptly lowered (below the steady-state extinction limit) to seek the purely kinetic extinction limit, as the flame responds quasi-steadily. Dilution on oxidizer side is employed to ensure that the flame size just before extinction is about double the radius of the burner to be utilized. Effect of starting flow rate duration time to establish a quasi-steady flame is evaluated with respect to how the flame responds to flow rate reduction and subsequent extinction. For radiative extinctions, H2/CH4/N2 and H2/CH4/He fuel mixtures are examined for different mole fractions compositions and issuing flow rates. Heat loss due to radiation is the dominant factor causing extinction as the flame spreads outwardly, when there is no steady solution for the given fuel and ambient mixtures. Extinction time varies less at high flow rates compared to low flow rates, for a given fuel mixture. High hydrogen concentration in the fuel mixture extends the extinction time, due to the high reactivity and diffusivity of hydrogen. For the same mole fraction of dilution in the fuel mixture and oxidizer mixture ambient, the one that is diluted by helium extinguishes sooner than the one contains nitrogen. Optically-thin models are shown to have shorter extinction times and bigger flame sizes than optically thick models due to more heat loss.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Dissertations and Theses	Xia T. "Computational Simulation of Spherical Diffusion Flame Dynamics and Extinction in Microgravity." M.S. Thesis, Mechanical & Aerospace Engineering, Rutgers University, 2015. , May-2015

Papers from Meeting Proceedings	Zhao P, Liang W, Deng S, Law CK, "On Premixed Cool Flames in the Counterflow" 9th U.S. National Combustion Meeting, Cincinnati, Ohio, May 17-20, 2015. 9th U.S. National Combustion Meeting, Cincinnati, Ohio, May 17-20, 2015. Paper 3E01. , May-2015